

The Borda Milan Pyramid and the VEHICLE Formula-as-Architecture

A Projection-Governed Framework for Complex Relational Systems

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Abstract

This paper presents the Borda Milan Pyramid as the architectural synthesis of the VEHICLE Formula-as-Architecture framework, a projection-governed approach for studying complex relational systems composed of internally structured nodes. The framework represents each node through an E.I.A.R.(V) state — orientation, information density, active coupling, relational compatibility, and verification-projection governance — and models global evolution through a dual-layer tension functional combining external relational discrepancy and internal incoherence. A projection-governed correction operator constrains node evolution toward admissible coherent regions, supporting simulation, regime diagnosis, and attractor classification. The paper introduces the layered architecture, a concrete quadratic computational instance, an initial operational taxonomy of attractor regimes, and a reproducibility path for future validation, while explicitly stating the current limits of the framework and the open problems required for deeper theoretical and empirical development.

Keywords: VEHICLE Systems Lab; Borda Milan Pyramid; Formula-as-Architecture; projection-governed systems; complex relational systems; graph dynamics; internal coherence; relational tension; attractor discovery; structured nodes; E.I.A.R.(V).

1. Foundational Positioning

VEHICLE Systems Lab emerges from a fundamental research impulse: to explore, understand, organize, and extend the frontier of knowledge. Its purpose is not to replace established scientific foundations, but to reorganize concepts from mathematics, systems theory, network science, optimization, computation, engineering, and dynamical modeling into a coherent architecture for studying tension, coherence, perturbation, transformation, and emergent regimes in complex relational systems.

The Borda Milan Pyramid and the VEHICLE Formula-as-Architecture stand upon a chain of prior scientific work developed by generations of investigators. Their contributions made possible the

concepts, tools, and formal languages that this framework now attempts to integrate. In this sense, VEHICLE is presented as an architectural continuation: a structured attempt to connect existing foundations into a projection-governed relational framework that can be tested, criticized, extended, and improved.

This work continues the documented research line of VEHICLE Systems Lab / AIMTG and should be read as part of an evolving research program connecting formal relational dynamics, E.I.A.R.(V) governance, dual-layer tension, and attractor discovery.

The central contribution of VEHICLE lies in architectural integration. Graph representation, structured node states, tension functionals, projection operators, local correction, regime classification, and scientific decision support are brought into one common structure in which internal coherence and external relational tension are studied together.

2. Need for an Architectural Synthesis

Many complex systems are difficult to study because instability does not arise from a single variable. Tension may appear between components, inside components, across scales, or through delayed propagation after perturbation. A system may appear stable externally while accumulating incoherence internally. Conversely, a system may display local disturbance while preserving global resilience.

Classical models of consensus, diffusion, optimization, and network dynamics provide essential tools for studying interaction and convergence. However, many real systems require a framework in which each node is treated not merely as a scalar value or passive vector point, but as an internally structured relational unit.

VEHICLE addresses this need by introducing a layered architecture in which:

1. an observed system is converted into a graph;
2. each node is represented as an internally structured state;
3. external tension and internal incoherence are measured together;
4. correction is governed by projection into an admissible coherent region;
5. operational attractor regimes are identified;
6. additional attractor classes may be discovered through simulation and domain-specific validation.

The purpose is to provide a common architecture for studying systems in which structure, relation, perturbation, coherence, and transformation evolve together.

3. The Borda Milan Pyramid

The Borda Milan Pyramid is the architectural representation of the VEHICLE framework. It is a layered synthesis of the research process by which an observed complex system is transformed into a projection-governed relational model.

Layer 1 — Observed Scientific Domain

The base of the pyramid begins with a real domain of study: planetary geology, orbital debris, environmental risk, resilient infrastructure, biological organization, multi-agent artificial intelligence, social systems, or any other complex system where structured components interact under tension.

Layer 2 — Relational Conversion

The observed system is converted into a graph:

$$G = (N, E),$$

where N is the set of nodes and E is the set of relational edges. Nodes represent structured entities, regions, agents, components, or units of analysis. Edges represent interaction, adjacency, influence, dependency, compatibility, or relational exposure.

Layer 3 — Structured VEHICLE Node

Each node is represented as an internally structured state:

$$S_i(t) = (E_i(t), I_i(t), A_i(t), R_i(t), V_i(t))^T \in \mathbb{R}^5.$$

The components are interpreted as follows:

- E_i : structural orientation or spin;
- I_i : information density or local memory;
- A_i : active internal coupling;
- R_i : relational compatibility with neighbors;
- V_i : verification-projection governance.

The node is therefore not a passive point. It is a component-structured relational unit.

Layer 4 — Dual-Layer Tension

The total tension of the system is modeled as the combination of external relational tension and internal coherence tension:

$$T(X) = T_{\text{ext}}(X) + T_{\text{int}}(X).$$

External tension measures discrepancy between connected nodes. Internal tension measures deviation from coherent structure within each node.

Layer 5 — Projection-Governed Correction

The system evolves through a projection-governed operator. A node first moves according to a descent-like correction, and the resulting candidate state is projected into an admissible coherent region K . This layer is central: VEHICLE does not allow evolution without constraint; it introduces correction under admissibility.

Layer 6 — Operational Attractor Regimes

The initial attractor classes A0–A6 are used as an operational taxonomy for simulation and analysis. They are not presented as an exhaustive catalogue of all possible relational regimes. They serve as an initial diagnostic language for observing stabilization, review, recovery, expansion, filtering, rigidity, and fluidity.

Layer 7 — Attractor Discovery

A central research frontier of VEHICLE is the discovery of additional attractor classes. When the architecture is applied to real scientific data, larger graphs, heterogeneous node structures, and domain-specific perturbations, new regimes may emerge beyond the initial A0–A6 taxonomy.

Layer 8 — Scientific Decision Support

At the upper level, the architecture supports scientific prioritization. It asks where tension accumulates, where coherence breaks, where perturbation propagates, where recovery is possible, where expansion becomes uncontrolled, and where further investigation or intervention should be directed.

4. Mathematical Core: A Concrete Quadratic Instance

This section presents a clean mathematical instance of the VEHICLE architecture suitable for computation, simulation, and reproducibility. It is not claimed to be the only possible implementation of VEHICLE.

Let $G = (N, E)$ be a finite undirected graph with $|N| = n$. Each node $i \in N$ carries a state:

$$S_i(t) = (E_i(t), I_i(t), A_i(t), R_i(t), V_i(t))^T \in \mathbb{R}^5.$$

The global state is:

$$X(t) = (S_1(t), S_2(t), \dots, S_n(t)) \in \mathbb{R}^{5n}.$$

4.1 Centering Operator

Let $\mathbf{1} \in \mathbb{R}^5$ be the vector of ones. Define the centering operator:

$$P_- = I_5 - \frac{1}{5} \mathbf{1} \mathbf{1}^T.$$

For any $S \in \mathbb{R}^5$,

$$P_- S = S - \bar{S} \mathbf{1}, \quad \bar{S} = \frac{1}{5} \mathbf{1}^T S.$$

The operator P_- removes the average component of the internal state. It is symmetric and idempotent:

$$P_-^T = P_-, \quad P_-^2 = P_-.$$

If the components E, I, A, R, V have heterogeneous units or scales, they must be normalized before Euclidean norms are used.

4.2 Internal Incoherence

Define the internal incoherence measure:

$$O(S) = \|P_- S\|^2.$$

This quantity measures the deviation of the internal components from their common average structure.

4.3 Admissible Coherent Region

For a threshold $T_{\text{limit}} \geq 0$, define the admissible coherent region:

$$K = \{S \in \mathbb{R}^5 : O(S) \leq T_{\text{limit}}\}.$$

Since $O(S)$ is a positive semidefinite quadratic form, K is a closed and convex sublevel set.

4.4 Tension Functional

A concrete quadratic implementation of total tension is:

$$T(X) = \frac{1}{2} \sum_{(i,j) \in E} w_{ij} \|S_i - S_j\|^2 + \frac{\lambda}{2} \sum_{i \in N} \|P_- S_i\|^2,$$

where $w_{ij} > 0$ are edge weights and $\lambda > 0$ controls the strength of internal coherence coupling.

The first term measures external relational discrepancy. The second term measures internal incoherence within nodes.

4.5 Gradient

The gradient of T with respect to S_i is:

$$\nabla_{S_i} T(X) = \sum_{j: (i,j) \in E} w_{ij} (S_i - S_j) + \lambda P_- S_i.$$

This expression combines the graph interaction term with the internal coherence correction term.

4.6 Projection-Governed Update

Let P_K denote the orthogonal projection onto K :

$$P_K(z) = \arg \min_{s \in K} \|s - z\|^2.$$

The projection-governed operator is:

$$V_{\text{op}}(S_i; X) = P_K(S_i - \eta \nabla_{S_i} T(X)),$$

where $\eta > 0$ is a descent step.

The relaxed update rule is:

$$S_i(t+1) = (1 - \alpha)S_i(t) + \alpha V_{\text{op}}(S_i(t); X(t)), \quad 0 < \alpha \leq 1.$$

This update expresses the core idea of VEHICLE: relational correction followed by admissibility-governed projection.

4.7 Local Stability Remark

Under standard regularity assumptions, if the gradient is locally Lipschitz, K is convex, P_K is the orthogonal projection onto K , and the step size η is sufficiently small, the projected update can be studied as a local descent mechanism. The present manuscript does not claim global convergence in full generality. Global stability requires additional assumptions and is left as an open theoretical problem.

This limitation is intentional: the framework is presented as an architecture with a concrete computational instance, not as a completed universal convergence theorem.

5. Architectural Novelty

The novelty of VEHICLE lies in its architectural integration. The individual elements belong to recognized scientific traditions and are organized within a common projection-governed structure.

VEHICLE brings together:

- internally structured five-dimensional nodes;
- external relational tension;
- internal coherence tension;
- projection-governed correction;
- operational attractor classification;
- controlled expansion and transformation;
- future attractor discovery across real scientific domains.

This integration allows a complex system to be studied not only through external agreement or diffusion, but through the simultaneous evaluation of internal admissibility, relational compatibility, perturbation response, and structural transformation.

6. Operational Attractor Taxonomy and Discovery

The A0–A6 attractor classes are introduced as an initial operational taxonomy. They are intended to support simulation, classification, and comparison between regimes. They should not be understood as an exhaustive list of all possible relational states.

A future operational implementation should define each attractor class through measurable quantities such as:

- internal incoherence $O(S_i)$;
- global tension $T(X)$;
- node displacement $\|S_i(t+1) - S_i(t)\|$;
- persistence over time;

- projection activation frequency;
- local versus global tension distribution;
- recovery, expansion, rigidity, or filtering behavior after perturbation.

The research program remains open to attractor discovery. One of the central objectives of future VEHICLE development is to determine whether additional attractor classes emerge under real data, larger graph scales, heterogeneous node types, and domain-specific perturbations.

This makes attractor discovery not an accessory element, but a primary research frontier of the architecture.

7. Reproducibility Path

To support scientific review and extension, future releases of the framework should include a reproducible computational package containing:

- source code for the tension functional and update operator;
- a documented implementation of P_- , K , and P_K ;
- synthetic graph examples;
- fixed random seeds;
- notebooks or scripts reproducing figures;
- parameter tables;
- CSV output examples;
- unit tests for gradient and projection functions;
- citation metadata and license information.

This reproducibility layer does not replace the architecture. It strengthens it by making the mathematical instance testable and auditable.

8. Limitations and Open Problems

The present version of the Borda Milan Pyramid should be interpreted as an architectural and computational framework under development. Its current limitations include:

1. **Operational calibration:** E.I.A.R.(V) components require domain-specific measurement rules.
2. **Attractor thresholds:** A0–A6 classes require quantitative thresholds for reproducible classification.
3. **Global convergence:** The manuscript does not claim a general global convergence theorem.
4. **Projection assumptions:** Guarantees depend on the convexity of K and properties of the projection operator.
5. **Empirical validation:** Real-domain validation remains necessary across multiple systems and scales.
6. **Scalability:** Large-scale behavior must be tested computationally and reported with time and memory metrics.
7. **Decision support:** Any use for prioritization or intervention requires careful domain interpretation and auditability.

These limitations do not weaken the architecture; they define the research path required for rigorous maturation.

9. Scientific Invitation

The Borda Milan Pyramid is proposed as an open architecture for investigation. Its purpose is to provide a structured way to observe complex systems, formalize relational tension, evaluate internal coherence, apply projection-governed correction, and study emergent attractor regimes.

VEHICLE Systems Lab invites scientific evaluation, replication, criticism, extension, and domain-specific testing. The framework is presented as a theory with a research architecture designed to be challenged, improved, and to continue evolving.

10. Preliminary Figure Plan

Recommended figures for the final manuscript:

1. **The Borda Milan Pyramid** — layered architecture from observed domain to scientific decision support.
 2. **VEHICLE Structured Node** — E.I.A.R.(V) components inside a node/cube.
 3. **Dual-Layer Tension** — external edge tension and internal node coherence.
 4. **Projection-Governed Update** — gradient correction followed by projection into K .
 5. **Attractor Discovery Layer** — A0–A6 as initial taxonomy with open discovery beyond the initial classes.
 6. **Reproducibility Pipeline** — manuscript, code, notebooks, DOI, GitHub, and website.
-

11. Conclusion

The Borda Milan Pyramid and the VEHICLE Formula-as-Architecture establish a projection-governed framework for studying complex relational systems through structured nodes, dual-layer tension, internal coherence, correction under admissibility, operational attractor regimes, and future attractor discovery.

The framework is proposed as both a mathematical and architectural contribution. Its central purpose is to organize existing scientific foundations into a coherent structure capable of supporting reproducible simulation, domain-specific validation, and deeper investigation into how complex systems accumulate tension, preserve coherence, respond to perturbation, transform, and generate emergent regimes.

VEHICLE Systems Lab presents this architecture as a foundation for future research, simulation, critique, and scientific collaboration across domains where relational complexity, structural transformation, and attractor discovery are central problems.

References

Borda Milan, R. (2026). *VEHICLE 3D with E.I.A.R.(V): A Projection-Governed Framework for Self-Stabilizing Relational Networks* (Version 1.0). Zenodo. <https://doi.org/10.5281/zenodo.19807591>

Appendix A — Symbol Table

Symbol	Meaning	Dimension / Type
$G = (N, E)$	Graph of the observed system	finite graph
N	Set of nodes	finite set
E	Set of edges	finite set
$S_i(t)$	State of node i at time t	$\mathbb{R}^5$
E_i	Orientation / spin	scalar
I_i	Information density	scalar
A_i	Active coupling	scalar
R_i	Relational compatibility	scalar
V_i	Verification-projection governance	scalar
$X(t)$	Global system state	$\mathbb{R}^{5n}$
w_{ij}	Edge weight	positive scalar
P_-	Centering operator	5×5 matrix
$O(S)$	Internal incoherence	scalar
K	Admissible coherent region	subset of $\mathbb{R}^5$
P_K	Projection onto K	operator
$T(X)$	Total tension	scalar
T_{ext}	External relational tension	scalar
T_{int}	Internal coherence tension	scalar
λ	Internal coherence weight	positive scalar
η	Descent step	positive scalar
α	Relaxation parameter	$(0, 1]$
V_{op}	Projection-governed operator	operator

Appendix B — Minimal Pseudocode

```
function compute_gradient(i, X, W, lambda):
    g = zero_vector(5)
    for j in neighbors(i):
        g += W[i,j] * (S_i - S_j)
    g += lambda * P_minus @ S_i
    return g

function project_K(z, P_minus, T_limit):
    if z.T @ P_minus @ z <= T_limit:
        return z
    solve min_s ||s-z||^2 subject to s.T @ P_minus @ s <= T_limit
    return projected_state

function V_op(i, X, W, lambda, eta, P_minus, T_limit):
    g = compute_gradient(i, X, W, lambda)
    z = S_i - eta * g
    return project_K(z, P_minus, T_limit)

function update_node(i, X, params):
    v = V_op(i, X, params)
    S_i_new = (1-alpha) * S_i + alpha * v
    return S_i_new
```

Appendix C — Recommended Repository Metadata

CITATION.cff

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message: "If you use this work or the associated software, please cite it as
below."
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authors:
  - family-names: "Borda Milan"
    given-names: "Roberto"
    orcid: "0009-0009-9047-1036"
version: "1.1"
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date-released: "2026-05-02"
url: "https://vehiclesystemslab.com"
keywords:
  - "VEHICLE Systems Lab"
  - "Borda Milan Pyramid"
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- "projection-governed systems"
- "complex relational systems"
- "attractor discovery"
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BibTeX

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